INCH-POUND

MIL-R-47025A (MI)
5 March 1991
SUPERSEDING
MIL-R-47025 (MI)
19 April 1974

MILITARY SPECIFICATION

RESIN, CASTING, FIRE RETARDANT, EPOXY BASE

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of fire retardant casting resin consisting of an epoxy resin and an amine type curing agent.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9330

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SPECIFICATIONS

MILITARY

MIL-I-16923 - Insulating Compound, Electrical, Embedding

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 149 - Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequences

ASTM D 150 - Standard Test Methods for A-C Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulating Materials

ASTM D 256 - Standard Test Methods for Impact Resistance of Plastics and Electrical Insulating Materials

ASTM D 257 - Standard Test Methods for D-C Resistance or Conductance of Insulating Materials

ASTM D 570 - Standard Test Method for Water Absorption of Plastics

ASTM D 696 - Standard Test Method for Coefficient of Linear Thermal Expansion of Plastics

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ASTM D 790	-	Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
ASTM D 792	-	Standard Test Methods for Specific Gravity (Relative Density) and Density of Plastics by Displacement
ASTM D 2240	-	Standard Test Method for Rubber Property-Durometer Hardness
ASTM D 2863	-	Standard Test Method For Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of Plastics (Oxygen Index)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

THE SOCIETY OF THE PLASTICS INDUSTRY

EPOXY RESIN FORMULATOR (ERF)

ERF-2-82	-	Method of Test for Gel Time and Peak Exothermic Temperature of Epoxy Compound
ERF-12-82	-	Method of Test for Linear Shrinkage of Epoxy Casting Resin During Cure

(Application for copies should be addressed to the Society of the Plastics Industry, Inc., 1275 K St. N.W., Suite 400, Washington, DC 20005.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article sample. Unless otherwise specified (see 6.2), a first article (see 4.2) of the casting resin shall meet the requirements of this specification.

3.2 Materials. The casting resin shall consist of an epoxy resin and an amine-type curing agent capable of being fully cured at a temperature not greater than 71.1° Celsius (C). The casting resin shall be black in color.

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3.3 Electrical properties. The casting resin shall meet the requirements specified in table I.

TABLE I. Electrical properties.

Property	Value
Dielectric strength, volts per mil at 20°C	350 minimum
Volume resistivity (ohm-centimeter) at 20°C	1×10^{12} minimum
Dielectric constant from 60 to 10^{10} cycles	6.0 maximum
Dissipation factor from 60 to 10^{10} cycles	0.09 maximum

3.4 Mechanical properties. The casting resin shall meet the requirements specified in table II.

TABLE II. Mechanical properties.

Property	Value
Flexural strength, pounds force per square inch (lbf/in^2)	8000 minimum
Izod impact, foot pounds per inch (lbf·in) of notch	0.2 minimum
Hardness (Shore D)	73 minimum
Coefficient of linear thermal expansion, inch per inch per 0.56°C	35×10^{-6} maximum
Coefficient of thermal conductivity British thermal units per hour per square foot per 0.56°C	2.7 minimum
Moisture absorption, percent (%)	1.0 maximum
Linear shrinkage (inch/inch)	0.01 maximum
Cure temperature including exothermic, °C	110 maximum

3.5 Physical properties.

3.5.1 Viscosity. The viscosity of the uncatalyzed casting resin shall be 15000 ± 5000 centipoises (cP).

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3.5.2 Specific gravity. The specific gravity of the casting resin at 23.9°C shall be 1.45 ± 0.15.

3.5.3 Shelf life. The shelf life of the epoxy resin and curing agent in unopened containers shall be not less than one year from date of manufacture when stored at temperature of not more than 27°C.

3.5.4 Fire retardant. The casting resin, when compounded, shall be self-extinguishing or non-flammable.

3.6 Workmanship. The workmanship shall be such as to insure a product which is uniform and in conformance with this specification. The resin shall be free of dirt, foreign material, or other contaminants.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 First article sample. The first article sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of casting resin. First article samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

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b. Quality conformance inspection (see 4.5 and table III).

4.4 First article inspection. First article inspection shall be conducted only on the first article sample and shall consist of all the examinations and tests specified herein.

4.5 Quality conformation inspection. Quality conformance inspection for acceptance of resin and curing agent shall be in conformance with table III.

TABLE III. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Viscosity	3.5.1	4.7.3.1
Specific gravity	3.5.2	4.7.3.2
Shelf life	3.5.3	4.7.3.3
Fire retardance	3.5.4	4.7.3.4

4.6 Lot size and sampling.

4.6.1 Lot size. Lot size shall consist of all the epoxy resin and curing agent manufactured at one time, from one batch, forming part of one contract or order, and submitted for acceptance at the same time and place.

4.6.2 Sampling. Unless otherwise specified (see 6.2), 1 pint of epoxy resin and 0.125 pint of curing agent from one unit representative of each lot (see 4.6.1) shall be selected at random for quality conformance testing. Each container of epoxy resin and curing agent shall be considered as a unit of product. Failure of sample to meet the quality conformance requirements shall be cause for lot rejection.

4.7 Test methods.

4.7.1 Electrical property tests.

4.7.1.1 Dielectric strength. The dielectric strength shall be determined in accordance with ASTM D149. The dielectric strength shall be in conformance with section 3.3 (see table I).

4.7.1.2 Volume resistivity. The volume resistivity shall be determined in accordance with ASTM D257. The dielectric constant shall be in conformance with section 3.3 (see table I).

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4.7.1.3 Dielectric constant. The dielectric constant shall be determined in accordance with ASTM D149. The dielectric constant shall be in conformance with section 3.3 (see table I).

4.7.1.4 Dissipation factor. The dissipation factor shall be determined in accordance with ASTM D150. The dissipation factor shall be in conformance with section 3.3 (see table I).

4.7.2 Mechanical property tests.

4.7.2.1 Flexural strength. The flexural strength shall be determined in accordance with ASTM D790. The flexural strength shall be in conformance with section 3.4 (see table II).

4.7.2.2 Izod impact. The impact resistance shall be determined in accordance with ASTM 256. The Izod impact resistance shall be in conformance with section 3.4 (see table II).

4.7.2.3 Hardness. The hardness shall be determined in accordance with ASTM D2240. The hardness shall be in conformance with 3.4 (see table II).

4.7.2.4 Coefficient of linear thermal expansion. The coefficient of linear thermal expansion shall be determined in accordance with ASTM D696. The coefficient of linear thermal expansion shall be in conformance with 3.4 (see table II).

4.7.2.5 Coefficient of thermal conductivity. The coefficient of thermal conductivity shall be determined in accordance with MIL-I-16923. The coefficient of thermal conductivity shall be in conformance with 3.4 (see table II).

4.7.2.6 Moisture absorption. The moisture absorption shall be determined in accordance with ASTM D570. The moisture absorption shall be in conformance with 3.4 (see table II).

4.7.2.7 Linear shrinkage. The linear shrinkage during the curing process shall be in accordance with ERF 12-82. The linear shrinkage shall be in conformance with 3.4 (see table II).

4.7.2.8 Peak exothermic temperature. The temperature during the entire curing cycle shall be measured at the geometric center of the test sample, in accordance with ERF 2-82, Method C, except that the constant temperature bath shall be employed so as to simulate the actual cure temperature-time cycle for the casting resin. The peak exothermic temperature shall be in conformance with 3.4 (see table II).

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4.7.3 Physical property tests.

4.7.3.1 Viscosity. The initial viscosity shall be determined with a Brookfield Viscosimeter with a number 6 spindle, or equal, rotating at a speed of 20 ± 5 revolutions per minute. The temperature of the casting resin during the test shall be at 23 ± 3°C. The viscosity shall be in conformance with 3.5.1.

4.7.3.2 Specific gravity. The specific gravity shall be determined in accordance with ASTM D792. The specific gravity shall be in conformance with 3.5.2.

4.7.3.3 Shelf life. The resin and curing agent shall be tested for conformance with all requirements specified herein after completion of the storage period (see 3.5.3). The manufacturer's certificate of compliance with shelf life requirements may be accepted during the storage period.

4.7.3.4 Fire retardance. The casting resin shall be self-extinguishing or non-flammable when tested in accordance with ASTM D 2863. Glass panels used during this test shall be prepared in accordance with Test Method 2021, FED-STD-141.

5. PACKAGING

5.1 Preservation, packaging, and packing. Unless otherwise specified in the purchase order, the epoxy resin may be furnished in tin plate cans or pails with metal lids. The curing agent shall be furnished in glass bottles. The containers shall be non-reactive to the epoxy resin and curing agent. Size of the container shall be specified by the procuring activity. The epoxy resin and curing agent shall be packaged in such a manner that damage during shipment and storage is prevented.

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The casting resin covered by this specification is intended for use as a fire retardant encapsulant for welded electronic circuits.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

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b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. When first article is required (see 3.1 and 4.2)

d. Conditions for quality conformance inspection (see 4.3)

e. Sampling plan if other than specified (see 4.6)

f. Container size (see 5.1).

g. Any special marking requirements (see 5.2)

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Compound
Curing agent, amine type
Electrical encapsulation material

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing activity:
Army - MI

Review:
DLA-GS

Civilian Coordinating Activity:
GSA - FSS

Project No. 9330-A026

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-47025A (MI)	5 March 1991

3. DOCUMENT TITLE
RESIN, CASTING, FIRE RETARDANT, EPOXY BASE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE *(Include Area Code)* (1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS *(Include Zip Code)* ATTN: AMSMI-RD-SE-TD-ST	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office	

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